

PRAVEEN JEEVANANTHAM

Embedded Hardware Engineer - Research Scientist

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PROFESSIONAL SUMMARY

Detail-oriented Embedded Hardware Engineer and Research Scientist with hands-on experience in hardware design, PCB layout, firmware development, embedded Linux, and real-time embedded systems. Proficient in STM32 microcontrollers, serial communication protocols (SPI, UART, I2C), power electronics design, and signal integrity analysis. Experienced in FMCW mmWave radar configuration and parameter optimization using TI IWRL6432AOPEVM. Skilled in embedded Linux bring-up, GPIO/DIO interfacing, driver configuration, and shell scripting on Single Board Computers. Experienced in full hardware development lifecycle from schematic capture and PCB design (Siemens PADS) to Gerber file generation, bring-up, and system-level debugging.

TECHNICAL SKILLS

Embedded Systems and Firmware: STM32 (STM32F303E), ARM Cortex-M, TI IWRL6432AOPEVM, Embedded C, Bare-metal Programming, RTOS concepts, Interrupt Handling, Register-level Programming, Peripheral Drivers

Embedded Linux: Custom Linux OS, Venus SBC, OS Flashing, Device Driver Configuration, GPIO/DIO Interfacing (sysfs), Shell Scripting, Kernel Log Analysis, Hardware Bring-up

Radar and RF Systems: FMCW mmWave Radar, TI IWRL6432AOPEVM, Chirp Configuration, Range-Doppler Processing, Range FFT, Doppler FFT, CFAR, mmWave Studio, mmWave SDK, Point Cloud Data, Radar Parameter Optimization

Communication Protocols: SPI, UART, I2C, Serial Communication, Fiber Optic Communication, Fiber Bragg Grating (FBG)

Hardware Design and Tools: PCB Design and Layout, Siemens PADS, Schematic Capture, Gerber File Generation, Component Placement and Routing, DC Power Supply Design, EMI/EMC Filtering, Signal Integrity, Noise Reduction, Voltage Regulators, Hardware Debugging

Microcontrollers and Platforms: STM32, Arduino, TI mmWave SoC, Venus SBC, IoT-based Systems, Sensor Integration

Programming Languages: Embedded C, Python, Shell Scripting (Bash), HTML, CSS, JavaScript

OS and Version Control: Linux, Windows, Git, GitHub

PROFESSIONAL EXPERIENCE

Research Scientist, Embedded Hardware and Control Systems

SAMEER, R&D of DIT, Govt. of India 2024 - Present

- * Contributed to the design and development of a Fire Control System, defining hardware architecture, system requirements, component selection, and performing schematic-level troubleshooting to resolve technical issues.
- * Led PCB design and bring-up of power supply subsystems for an Integrated Control System, ensuring stable DC rail outputs, noise filtering, and efficient power management under operational loads.
- * Developed and validated serial communication firmware for the Integrated Control System, implementing SPI and UART protocols to ensure reliable, low-latency data transmission between embedded subsystems.
- * Performed hardware debugging and root cause analysis on PCB assemblies, utilizing oscilloscopes and multimeters to diagnose signal integrity, power, and communication issues.
- * Documented hardware design specifications, test procedures, and system requirements in alignment with embedded hardware development lifecycle standards.
- * Worked on Venus SBC (Single Board Computer) running a custom Linux OS, gaining hands-on experience with embedded Linux bring-up and board-level hardware integration.
- * Flashed custom OS images onto the Venus SBC and configured device drivers to enable peripheral communication and hardware functionality.
- * Interfaced with GPIO and DIO using Linux sysfs and kernel interfaces, enabling hardware control and sensor interfacing directly from the OS level.
- * Developed and executed shell scripts for automated hardware control, system initialization, and peripheral testing on the embedded Linux platform.
- * Debugged hardware-software interaction issues by analyzing kernel logs, checking driver initialization sequences, and validating signal behavior using test equipment.

Intern Railway Scientist, Fiber Optic Sensor Systems

Lab to Market Innovations Pvt Ltd, CSIR Taramani Campus 2022

- * Completed internship focused on Application of Sensors in Fiber Optic Communication, gaining hands-on experience with optical sensor systems and signal acquisition.
- * Developed Fiber Bragg Grating (FBG)-based structural health monitoring systems for railway infrastructure, involving sensor calibration, data acquisition, and signal analysis to detect structural strain and deformation.
- * Contributed to research aimed at improving railway safety and reliability through real-time embedded sensing and monitoring technology.

PROJECTS

Custom SPI Clock Signal Generation, STM32F303E - *Embedded C, STM32, ARM Cortex-M, SPI Protocol*

- * Designed and implemented bare-metal firmware for the STM32F303E microcontroller to generate a precise 30-cycle SPI clock signal, directly configuring GPIO and SPI peripheral registers without HAL abstraction.
- * Validated waveform accuracy and timing using oscilloscope measurements; optimized register-level configuration for minimal latency and deterministic behavior.

Custom DC Power Supply PCB Design - *Siemens PADS, PCB Layout, Power Electronics, EMI Filtering*

- * Designed a noise-free regulated DC power supply circuit using Siemens PADS, performing full PCB layout including schematic capture, component placement, trace routing, and Gerber file generation for manufacturing.
- * Implemented DC filtering stages to suppress ripple and conducted design review for EMI compliance and thermal management.

Smart Metering Plug, IoT Power Monitor - *Arduino, IoT, Embedded C, Sensor Integration*

- * Designed and built an affordable IoT-based smart metering plug to monitor real-time AC power consumption and measure battery state-of-charge for connected electric vehicles.
- * Integrated current/voltage sensors with an Arduino microcontroller, implementing embedded firmware for data acquisition, unit calculation, and serial data output.

E-Commerce Website - *HTML, CSS, JavaScript*

- * Built a responsive e-commerce frontend with client-side form validation, interactive UI components, and consistent CSS design system.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering (ECE)

KCG College of Technology | 2023 | CGPA: 7.08

Higher Secondary, Class XII

Anna Gem SPS | 2019 | 58.5%

Secondary School, Class X

Kendriya Vidhyalaya CLRI | 2017 | CGPA: 5.8